$1\mu A = 1 micro ampere = 10^{-6} ampere$

Therefore,

$$I_S = 10^6 \frac{div}{A} = 1\frac{div}{10^{-6}A} = 1\frac{div}{\mu A}$$

$$I_S = 1\, div(\mu A)^{-1}$$

(b) Voltage sensitivity

$$V_S = \frac{\theta}{V} = \frac{50 div}{25 mV} = 2\times10^3 div\, V^{-1}$$

(c) The resistance of the galvanometer is

$$R_g = \frac{I_S}{V_S} = \frac{10^6 \frac{div}{A}}{2\times10^3 \frac{div}{V}} = 0.5\times10^3 \frac{V}{A} = 0.5 k\Omega$$

EXAMPLE 3.30

The resistance of a moving coil galvanometer is made twice its original value in order to increase current sensitivity by 50%. Will the voltage sensitivity change? If so, by how much?.

Solution

Yes, voltage sensitivity will change.

Voltage sensitivity is $V_S = \frac{I_S}{R}$

When the resistance is doubled, then new resistance is $R' = 2R$

Increase in current sensitivity is $I_S' = \left(1+\frac{50}{100}\right) I_S = \frac{3}{2} I_S$

The new voltage sensitivity is

$$V_S' = \frac{\frac{3}{2} I_S}{2R} = \frac{3}{4} V_S$$

Hence the voltage sensitivity decreases. The percentage decrease in voltage sensitivity is

$$\frac{V_S - V_S'}{V_S} \times 100\% = 25\%$$

Conversion of galvanometer into ammeter and voltmeter

A galvanometer is very sensitive instrument to detect the current. It can be easily converted into ammeter and voltmeter.

Galvanometer to an Ammeter

Ammeter is an instrument used to measure current flowing in the electrical circuit. The ammeter must offer low resistance such that it will not change the current passing through it. So ammeter is connected in series to measure the circuit current.

A galvanometer is converted into an ammeter by connecting a low resistance in parallel with the galvanometer. This low resistance is called shunt resistance S. The scale is now calibrated in ampere and the range of ammeter depends on the values of the shunt resistance.

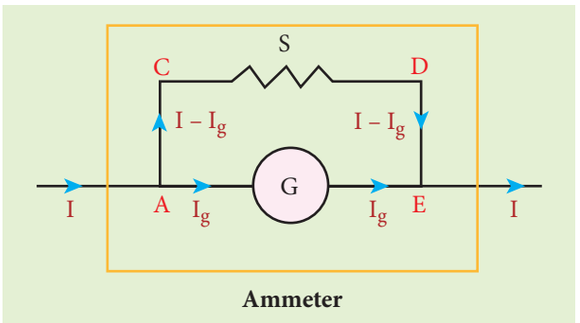


Figure 3.69 Shunt resistance connected in parallel

Let I be the current passing through the circuit as shown in Figure 3.69. When current I reaches the junction A, it divides into two components. Let I_g be the current passing through the galvanometer of resistance R_g through a path AGE and the remaining current $(I - I_g)$ passes along the path ACDE through shunt resistance S. The

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